5.Cloud Compute Services Overview

5.1 Amazon EC2 (Virtual servers in the cloud)

5.2 Amazon EC2 Auto Scaling (Scale compute capacity to meet demand)

5.3 Amazon LightSail (Launch and manage virtual private servers)

5.4 AWS Elastic Beanstalk (Run and manage web apps)

5.5 AWS Lambda (Run code without thinking about servers).

5.6 Case Study on Cloud Services.

5.1 Amazon EC2 (Virtual servers in the cloud)

Amazon Elastic Compute Cloud (Amazon EC2) is a cloud service provided by Amazon Web Services (AWS) that offers scalable and flexible virtual servers. These virtual servers, called instances, allow users to run applications in the cloud without having to invest in expensive physical hardware.

Think of Amazon EC2 as a computer rental service on the internet. Instead of buying a physical computer (server), you can rent a virtual one from AWS and use it for as long as you need. You can choose the type of virtual server based on how much power (CPU), memory (RAM), and storage you need.

Example: Imagine you are a software developer and want to **host a website**. Instead of buying a physical server and managing it, you can:

1. **Create an EC2 instance** – This is like turning on a virtual computer.
2. **Install a web server (e.g., Apache, Nginx, or IIS)** – This helps run your website.
3. **Deploy your website files** – You upload and run your website.
4. **Make it available to users** – People can access your site via the internet.

When you don't need the server anymore, you can **stop or delete the EC2 instance**, and you will only be charged for the time you used it.

Key Features of Amazon EC2

- ➢ **On-Demand Instances:** You can create instances whenever needed and pay only for the time you use them.
- ➢ **Scalability (Auto Scaling) :** EC2 allows automatic scaling, which means it can **increase or decrease** the number of servers based on demand.
- ➢ **Multiple Instance Types :** EC2 provides different types of instances optimized for computing, memory, storage, or networking needs. Examples:
  - **T-Series (General Purpose):** Good for websites and small applications.
  - **C-Series (Compute-Optimized):** Ideal for high-performance applications.
  - **R-Series (Memory-Optimized):** Best for databases and big data applications.
- ➢ **Security & Reliability:** EC2 instances are protected by **AWS security features**, including firewalls, identity management, and encryption.
- ➢ **Integration with AWS Services :** EC2 works with other AWS services like:

- **Amazon S3** (Storage)
- **Amazon RDS** (Databases)
- **AWS IAM** (Security & Access Management)

Key Benefits of EC2

- ➢ **Scalable** – You can increase or decrease resources based on demand.
- ➢ **Cost-Effective** – You only pay for what you use (Pay-as-you-go model).
- ➢ **Flexible** – You can choose different operating systems like Windows, Linux, etc.
- ➢ **Secure** – AWS provides security features to protect your data.

5.2 Amazon EC2 Auto Scaling (Scale compute capacity to meet demand)

Amazon EC2 Auto Scaling is a service that **automatically adjusts** the number of EC2 instances in your application based on demand. It ensures that you always have the right number of instances running—**increasing capacity during high traffic** and **reducing instances when demand is low** to save costs.

How It Works:

1. **Monitor Demand** – Auto Scaling tracks the usage of your application (CPU, memory, network traffic, etc.).
2. **Scaling Up (Increase Capacity)** – When more users visit your application, EC2 Auto Scaling adds more instances to handle the load.
3. **Scaling Down (Reduce Capacity)** – When traffic decreases, it removes unnecessary instances to save costs.

Key Benefits:

- ➢ **High Availability** – Ensures your application remains responsive even during traffic spikes.
- ➢ **Cost Optimization** – Avoids paying for unused capacity by automatically removing idle instances.
- ➢ **Automation & Efficiency** – No manual intervention is needed; Auto Scaling works in real-time.

Example :Imagine you run an online store. During normal hours, you have a few customers, but during sales or festive seasons, thousands of people visit your store at the same time. If your website doesn't have enough servers to handle the traffic, it may slow down or crash.

Amazon EC2 Auto Scaling helps solve this problem by automatically adding more servers (instances) when demand increases and reducing them when demand drops.

Types of Scaling in Amazon EC2 Auto Scaling

1. **Dynamic Scaling** – Adjusts capacity automatically based on real-time metrics and policies.
2. **Predictive Scaling** – Uses machine learning to anticipate future demand and scale accordingly.
3. **Scheduled Scaling** – Allows you to set specific times for scaling (e.g., scale up every weekday at 9 AM and scale down at 6 PM).

Use Cases of Amazon EC2 Auto Scaling

- **E-commerce websites** – Handles sudden spikes during sales or holiday seasons.
- **Media & Streaming Services** – Manages fluctuating viewership demands.
- **SaaS Applications** – Ensures performance consistency for cloud-based apps.
- **Gaming Platforms** – Dynamically scales game servers based on player activity.
- **Financial & Banking Systems** – Maintains stability even during stock market surges.

5.3 Amazon LightSail (Launch and manage virtual private servers)

Amazon LightSail is a cloud service from AWS that makes it easy to launch and manage virtual private servers (VPS). It is designed for students, developers, small businesses, and people who are new to cloud computing. LightSail provides everything you need to build websites and web applications at a low cost. A VPS is a virtual machine (computer) that runs in the cloud. It works like a regular computer but is hosted online. You can install software, host websites, run applications, and store files.

Features of Amazon LightSail:

1. **Easy to Set Up** – Launch a server in just a few clicks.
2. **Pre-configured Apps** – Includes WordPress, Magento, Joomla, etc.
3. **Fixed Pricing** – Simple monthly pricing with no hidden costs.
4. **Built-in Networking** – Includes static IP, DNS management, and firewall.
5. **Snapshots** – Take backups of your server.

6. **SSH Access** – Securely connect to your VPS using terminal/command line.

What Can You Do with LightSail?

- Host websites and blogs.
- Run e-commerce platforms.
- Develop and test applications.
- Host learning platforms or projects.
- Set up file storage or game servers.

Why Use Amazon LightSail?

- Beginner-friendly alternative to Amazon EC2.
- Easy to use compared to other complex cloud services.
- Ideal for students and developers who want to **experiment, learn, or build small projects**.
- Offers a **free trial** for new users.

Example: Launching a WordPress Website with LightSail

1. **Login** to your AWS account and open LightSail.
2. Click **"Create Instance"**.
3. Choose **Instance Location** (region).
4. Select **"Linux/Unix"** and choose **WordPress** from the application blueprint.
5. Choose your **instance plan** (e.g., $3.50/month).
6. Name your instance and click **Create Instance**.
7. After launch, LightSail provides a **public IP**—you can access your website by typing it into a browser.
8. Login to your WordPress admin dashboard using the default credentials provided.

5.4 AWS Elastic Beanstalk (Run and manage web apps)

AWS Elastic Beanstalk is a **Platform as a Service (PaaS)** offered by Amazon Web Services that enables developers to deploy, manage, and scale web applications and services automatically.

AWS Elastic Beanstalk is a tool that helps you run your website or web application on the internet **without setting up servers yourself**. You just upload your code, and it takes care of everything else like launching servers, scaling them, and keeping the site healthy.

Key Features:

- ➢ **Easy Deployment:** Developers can simply upload their application code, and Elastic Beanstalk automatically handles the deployment – from capacity provisioning, load balancing, and auto-scaling to application health monitoring.
- ➢ **Supports Multiple Languages:** Includes support for Java, .NET, PHP, Node.js, Python, Ruby, Go, and Docker.
- ➢ **Managed Infrastructure:** Manages the underlying infrastructure like EC2 instances, load balancers, and security groups.
- ➢ **Auto Scaling & Load Balancing:** Automatically scales the application based on demand and uses Elastic Load Balancer for distributing traffic.
- ➢ **Monitoring:** Integrated with Amazon CloudWatch for real-time performance monitoring.
- ➢ **Customization:** Provides configuration options for more control over the environment (via configuration files or the management console).

Benefits:

- ➢ **No Infrastructure Management:** Focus on writing code rather than managing infrastructure.
- ➢ **Quick Deployment:** Rapidly deploy applications with minimal configuration.
- ➢ **Cost-effective:** Pay only for the AWS resources used.
- ➢ **Environment Management:** Easily create different environments for development, testing, and production.

Example :

Let’s say you made a small website using **Python (Flask)** on your computer.
Now you want to put that website on the internet so anyone can use it.

1. **You zip your project files** (your website code).
2. **Go to the AWS Elastic Beanstalk dashboard**.
3. **Upload your zip file** and choose the platform (e.g., Python).
4. **Click "Launch"** – That’s it!

Elastic Beanstalk will:

- ➢ Create the necessary **servers (EC2)** to run your site.
- ➢ Set up a **load balancer** to handle traffic.
- ➢ **Monitor** your app to make sure it's running.
- ➢ **Auto-scale** if more people start visiting your website.

You don't need to learn how to manage servers – Beanstalk does it for you

5.5 AWS Lambda (Run code without thinking about servers).

AWS Lambda is a serverless compute service offered by Amazon Web Services (AWS) that allows you to run code without the need to provision, manage, or scale servers manually. With Lambda, you simply upload your code, and the service automatically handles everything required to run and scale it based on demand. This makes it an ideal solution for applications that require flexible scaling, event-driven execution, and cost-efficiency, as you're only charged for the actual compute time your code consumes.

Lambda supports multiple programming languages like Python, Node.js, Java, and more, and integrates seamlessly with other AWS services such as S3, DynamoDB, API Gateway, and CloudWatch. It is widely used in scenarios like real-time file or data processing, backend services for web and mobile apps, and automation tasks. By eliminating the need for server management, AWS Lambda enables developers to focus solely on writing and deploying code quickly, making it a cornerstone of modern, cloud-native application development.

Key Features:

- **Serverless**: No need to manage infrastructure. AWS handles provisioning, scaling, and availability.
- **Event-driven**: Lambda functions can be triggered by AWS services like S3, DynamoDB, API Gateway, or custom events.
- **Automatic scaling**: Lambda automatically scales the number of instances based on the number of incoming requests.
- **Pay-per-use**: You are charged only for the compute time your code uses, in increments of milliseconds.

Use Cases:

- Real-time file processing (e.g., image or video upload to S3)

- Backend for web or mobile apps
- Real-time data streaming using services like Kinesis
- Automation and scheduled tasks (using Amazon EventBridge or CloudWatch)

How it works:

- You write a small piece of code (called a **function**)
- You set a **trigger** (like file upload, API request, etc.)
- Whenever the event happens, Lambda **runs your code automatically**

Benefits:

- No server management
- Efficient cost model
- High scalability
- Tight integration with AWS ecosystem

Difference between Amazon Lightsail, Elastic Beanstalk, and AWS Lambda

Feature	Amazon Lightsail	AWS Elastic Beanstalk	AWS Lambda
Type	Simplified VPS (IaaS)	Platform as a Service (PaaS)	Serverless (FaaS)Function as service
Main Use	Small websites, simple apps	Deploy & manage web apps	Run small functions/events
Setup	Very easy (pre-configured)	Easy (upload code)	No setup needed
Infrastructure Control	Limited control	Moderate control	No control (fully managed)
Scaling	Manual / limited	Auto scaling supported	Automatic scaling
Execution	Runs continuously	Runs continuously	Runs only when triggered
Pricing	Fixed monthly cost	Pay for AWS resources used	Pay per execution/time
Best For	Beginners, small projec	Medium-large applications	Event-driven tasks, APIs